

SET 17

1. R: Create a 4x4 marks matrix and calculate row totals, column totals and overall mean.

The screenshot shows the RStudio interface with the following code in the console:

```
R - R4.6.1 ~\...
Platform: x86_64-w64-mingw32/x64

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You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from ~\.RData]

> # Create a 4x4 marks matrix
> marks <- matrix(c(
+   80, 75, 90, 85,
+   70, 85, 78, 88,
+   92, 89, 95, 90,
+   65, 72, 68, 75
+ ), nrow = 4, byrow = TRUE)
>
> # Display the matrix
> print(marks)
     [,1] [,2] [,3] [,4]
[1,]  80   75   90   85
[2,]  70   85   78   88
[3,]  92   89   95   90
[4,]  65   72   68   75
>
> # Row totals
> row_totals <- rowSums(marks)
> print(row_totals)
[1] 330 321 366 280
>
> # Column totals
> column_totals <- colSums(marks)
> print(column_totals)
[1] 307 321 331 338
>
> # Overall mean
> overall_mean <- mean(marks)
> print(overall_mean)
[1] 81.0625
```

The Environment pane on the right shows the following objects:

Object	Class	Dimensions	Values
marks	num	[1:4, 1:4]	80 70 92 65 75 85 89 72 90 78 ...
row_totals	num	[1:4]	307 321 331 338
column_totals	num	[1:4]	81.0625
overall_mean	num	[1:4]	330 321 366 280

2. R: Calculate mean absolute deviation from the median for 9,12,15,18,20,24,28,31.

The screenshot shows the RStudio interface with the following code in the console:

```
R - R4.6.1 ~\...
> # Given data
> x <- c(9, 12, 15, 18, 20, 24, 28, 31)
>
> # Calculate median
> median_x <- median(x)
>
> # Calculate Mean Absolute Deviation from median
> MAD <- mean(abs(x - median_x))
>
> # Display results
> print(paste("Median =", median_x))
[1] "Median = 19"
> print(paste("Mean Absolute Deviation =", MAD))
[1] "Mean Absolute Deviation = 6.125"
```

The Environment pane on the right shows the following objects:

Object	Class	Dimensions	Values
MAD	num		6.125
median_x	num		19
x	num	[1:8]	9 12 15 18 20 24 28 31

3. WEKA: Create credit-approval ARFF data (Income, Debt, EmploymentStatus, Approved); apply Naive Bayes and record evaluation.

Dataset: Credit Approval

```
@relation credit_approval
@attribute Income numeric
@attribute Debt numeric
@attribute EmploymentStatus {Employed,SelfEmployed,Unemployed}
@attribute Approved {Yes,No}
@data
25000,12000,Employed,No
35000,10000,Employed,Yes
45000,8000,Employed,Yes
30000,18000,SelfEmployed,No
55000,9000,SelfEmployed,Yes
22000,15000,Unemployed,No
48000,7000,Employed,Yes
38000,11000,SelfEmployed,Yes
```

The screenshot shows the Weka Explorer interface with the Naive Bayes classifier selected. The 'Test options' section shows 'Cross-validation' with 8 folds. The 'Classifier output' section displays the following results:

Classifier output	SelfEmployed	Unemployed	[total]
SelfEmployed	3.0	2.0	
Unemployed	1.0	2.0	
[total]	8.0	6.0	

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Metric	Value	Percentage
Correctly Classified Instances	8	100 %
Incorrectly Classified Instances	0	0 %
Kappa statistic	1	
Mean absolute error	0.0096	
Root mean squared error	0.0218	
Relative absolute error	1.8212 %	
Root relative squared error	4.0396 %	
Total Number of Instances	8	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Yes
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	No
Weighted Avg.	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	

=== Confusion Matrix ===

```
a b <-- classified as
5 0 | a = Yes
0 3 | b = No
```

4. WEKA: Apply SMO to the credit data and compare ROC area and F1 with Naive Bayes.

Dataset: Credit Approval – SMO

Use the same Credit Approval ARFF dataset from Set 17 Q3.

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

ChooseSMO -C 1.0 -L 0.001 -P 1.0E-12 -N 0 -V -1 -W 1 -K "weka.classifiers.functions.supportVector.PolyKernel -E 1.0 -C 250007" -calibrator "weka.classifiers.functions.Logistic -R 1.0E-

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation

☐ Percentage split

Folds

%

8

66

Set...

More options...

(Nom) Approved

Start

Stop

Result list (right-click for options)

23:21:21 - bayes.NaiveBayes

23:22:56 - functions.SMO

Classifier output

SelfEmployed	3.0	2.0
Unemployed	1.0	2.0
[total]	8.0	6.0

Time taken to build model: 0 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	8	100	%
Incorrectly Classified Instances	0	0	%
Kappa statistic	1		
Mean absolute error	0.0096		
Root mean squared error	0.0218		
Relative absolute error	1.8212	%	
Root relative squared error	4.0396	%	
Total Number of Instances	8		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Yes
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	No
Weighted Avg.	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	

=== Confusion Matrix ===

a b <-- classified as

5 0 | a = Yes

0 3 | b = No

Status

OK

Log